

Vision-Language Model Guided Source-Free Domain Adaptation via Optimal Transport

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Abstract

Unsupervised domain adaptation transfers knowledge from a labeled source domain to an unlabeled target domain. When source data cannot be accessed, source-free domain adaptation (SFDA) becomes a practical alternative. However, existing SFDA methods mainly rely on pseudo-label based self-training, which often accumulates noise and bias under large domain gaps. We propose VSFOT, a framework that leverages a pretrained Vision-Language Model (VLM) to guide optimal transport (OT) alignment between target features and source prototypes. Instead of relying on unreliable pseudo-labels, VSFOT employs VLM-derived semantic priors and an OT-based matching strategy to achieve stable and reliable adaptation. To further enhance domain alignment, VSFOT incorporates a bidirectional distillation mechanism in which the model learns semantic consistency from the VLM, while the VLM is refined using task-specific cues from the model. These two stages alternate during training. By combining the generalization ability of the VLM with the discriminative power of the task model, VSFOT achieves robust, source-free adaptation and consistently outperforms existing SFDA methods on four benchmark datasets. The code is available¹.

1. Introduction

Deep learning models often suffer from severe performance drops when applied to new domains, mainly due to domain shifts. Unsupervised domain adaptation (UDA) addresses this issue by leveraging labeled data from a source domain to enhance performance on an unlabeled target domain [23]. However, in sensitive fields like healthcare, autonomous driving, and edge systems, there are strict constraints on privacy, regulation, and storage. These limitations often render UDA inapplicable. Thus, source-free domain adaptation (SFDA) has emerged as a practical

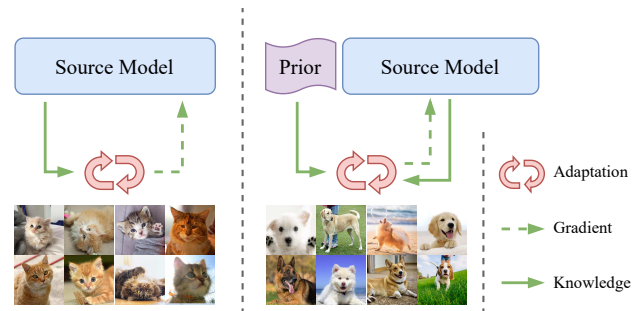


Figure 1. (Left) Traditional source-free domain adaptation perform adaptation using only a source model and unlabeled target data. (Right) Under the paradigm of external prior guidance, the proposed method leverages external priors as semantic references during adaptation, thereby improving performance.

and privacy-preserving alternative [12, 21]. Nevertheless, SFDA remains a highly promising yet inherently challenging paradigm, as adaptation must be performed without any access to source data.

To address the limitation that the source domain is inaccessible, a number of successful SFDA methods have been developed. However, most of these methods are fundamentally based on self-training, in which the model uses its own predictions as supervision signals to iteratively improve itself on the target data [12]. To optimize this self-training process, many techniques have been adopted, such as confidence thresholding [3], entropy minimization [15, 17], and curriculum learning [9, 37]. This line of research still fails to address the core issue of confirmation bias in SFDA. The supervisory signals generated during self-training tend to reinforce incorrect predictions over time, eventually leading to entrenched and irreversible biases. Therefore, developing a mechanism to effectively correct confirmation bias within SFDA is of great necessity.

In response, recent studies have explored using external semantic priors as guidance to assist the source model in adapting to the target domain distribution [6, 34, 35]. In this paradigm, the domain-invariant knowledge contained

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¹<https://github.com/TangXu-Group/VSFOT>

in external semantic priors can correct the model’s own prediction errors, thereby preventing incorrect predictions from being memorized. As shown in Figure 1, the difference between incorporating external priors and traditional SFDA methods is illustrated. During this prior-guided adaptation process, an intuitive approach is to employ semantic supervision, encouraging the source model to learn the target domain distribution and make more accurate and confident predictions [24, 40]. However, existing methods always employ a hard semantic supervision manner, which leads to degraded adaptation performance due to noise interference [39]. Therefore, it is essential to develop a prior-guided approach that reduces semantic supervision noise and maintains strong robustness.

Building on the above discussion, we propose VSFOT, a framework that leverages vision-language models (VLMs) as external semantic priors to provide stable and domain-invariant guidance during adaptation. Following the idea of reducing semantic supervision noise, VSFOT reformulates the adaptation process as a semantic optimal transport (OT) problem [27], where OT performs a soft alignment strategy that avoids the noise introduced by hard labels. Moreover, benefiting from its geometry-aware nature, OT provides strong robustness when aligning complex distributions [22]. To implement this, the semantic priors from VLMs are injected into the SFDA process by optimizing an OT-based distribution alignment objective, achieving semantically meaningful adaptation without relying on noisy pseudo-labels. Furthermore, since VLMs lack task-specific knowledge, a reverse knowledge transfer path is introduced. By encouraging the model to align high-confidence class distributions, task-relevant cues are extracted to refine the VLM’s guidance and enhance adaptation performance. Finally, these two directions are alternately optimized, forming a bidirectional distillation process that enables VLMs to more effectively guide the source model and achieve stable adaptation to the target domain.

Our main contributions are summarized as follows:

- A bidirectional distillation scheme is proposed to effectively introduce prior knowledge, enabling interaction and iterative optimization between the source and external models to enhance adaptation performance.
- We show that reformulating adaptation as an OT problem effectively mitigates noise interference in semantic supervision and enhances the robustness of predictions.
- Across four benchmarks, results validate our method’s effectiveness and demonstrate strong competitiveness.

2. Related Work

2.1. Source-Free Domain Adaptation

SFDA addresses the challenge of adapting a pretrained source model to an unlabeled target domain when the source

data is inaccessible. Existing approaches can be broadly grouped into model-driven and data-driven paradigms [12]. Model-driven methods directly adapt the source model using target data through techniques, such as pseudo-labeling [20, 28], contrastive learning [17, 45], consistency regularization [19, 33], and attention mechanisms [13, 31]. In contrast, data-driven approaches aim to recover implicit source knowledge from model parameters. They often construct class prototypes from classifier weights or predictions and align target features through clustering or prototype matching [41, 42, 44]. While these strategies achieve remarkable progress, they depend solely on information within the source model and unlabeled target data, which limits their ability to handle large domain shifts. Recent works attempt to inject external semantic priors to stabilize adaptation, motivating the exploration of VLMs as a new form of transferable knowledge.

2.2. Vision-Language Model in Domain Adaptation

VLMs, pretrained on large-scale image–text pairs, demonstrate strong zero-shot recognition and cross-domain generalization capabilities. Their use in domain adaptation has expanded rapidly. Early studies focus on parameter-efficient tuning, such as prompt learning and adapter insertion, to adapt VLMs to new domains [7, 10, 32]. Other works retain frozen VLMs and design inference-time strategies, including prompt calibration and sample reweighting [16, 47]. These approaches inspired a wave of VLM-guided domain adaptation, where VLMs provide semantic supervision for target models. In SFDA, this idea is often realized by generating pseudo-labels from VLMs predictions to guide target model updates [35, 36]. However, pseudo-label-based supervision remains fragile when the domain discrepancy is large, as incorrect labels can propagate bias and degrade adaptation quality. This challenge calls for approaches that leverage VLMs without relying on pseudo-labels, leading to the integration of distributional or semantic alignment techniques.

2.3. Optimal Transport for Cross-Domain Alignment

OT provides a principled framework for aligning distributions while preserving structural correspondence [4]. It has been successfully applied to UDA and SFDA tasks for feature matching and prototype alignment [14, 18, 42, 46]. Recent extensions combine OT with multimodal foundation models to integrate semantic cues into the transport process [1, 47]. These works highlight the potential of coupling OT with VLMs to bridge visual and semantic spaces. Yet in strictly source-free SFDA, OT-only self-training can amplify pseudo-label errors, while VLM-only supervision is often noisy under large shifts. We therefore embed priors from VLMs into the OT cost and marginals, exploring a

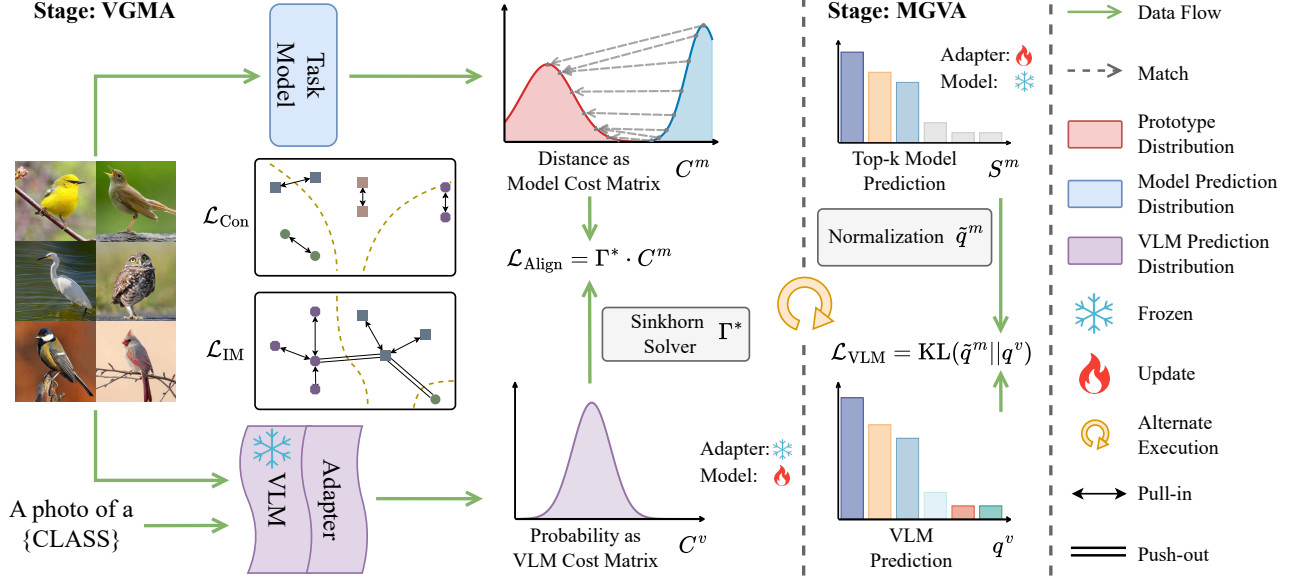


Figure 2. Illustration of the vision-language model (VLM) guided source-free domain adaptation via optimal transport. (Left) In the VGMA stage, the VLM guides the task model to adapt to the target domain. Contrastive loss and information maximization are incorporated to further enhance the model’s adaptation. (Middle) In the MGVA stage, high-confidence model predictions are used to refine VLMs, injecting task-specific knowledge and enhancing its guidance capability. (Right) The legend shows key symbols and visual cues used in the illustration.

more robust approach to domain adaptation.

3. Method

3.1. Problem Definition

Before introducing VSFOT, we first formalize the problem setting. Let the source domain be denoted as $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, which contains N_s labeled samples from K classes. The target domain is represented as $\mathcal{D}_t = \{x_i^t\}_{i=1}^{N_t}$, consisting of N_t unlabeled samples. The source model, also referred to as the task model, $\mathcal{G} = (f, c)$ consists of a feature extractor f and a classifier c , which are pretrained on the labeled source domain $\mathcal{D}_s$. In the SFDA setting, access to the source data and target labels is prohibited during adaptation. The objective is to adapt $\mathcal{G}$ to the unlabeled target domain $\mathcal{D}_t$, where an auxiliary VLM $\mathcal{V}$ provides semantic priors to guide cross-domain alignment.

3.2. Overview

VSFOT leverages the semantic prior knowledge from VLMs to facilitate SFDA via a bidirectional distillation mechanism. As illustrated in Figure 2, the overall framework alternates between two complementary stages: VLM-Guided Model Alignment (VGMA) and Model-Guided VLM Adaptation (MGVA). In VGMA, classifier weights trained on the source domain serve as class prototypes. The target features are aligned to these prototypes via an OT ob-

jective, in which semantic priors from VLMs are incorporated into the computation of the transport plan. This design encourages semantically meaningful alignments and mitigates the errors introduced by unreliable pseudo-labels. Auxiliary contrastive and information maximization regularizers are jointly applied to enhance representation discriminability and promote balanced predictions. In MGVA, high-confidence predictions from the task model act as anchors to adapt a lightweight module within VLMs, enabling them to capture task-specific cues in the target domain. This adaptation enhances a VLM’s domain awareness and strengthens its ability to guide subsequent alignments. These two stages alternate iteratively, where VLMs refine distribution alignment and the adapted task model enhances a VLM’s task awareness, forming a mutually reinforcing loop for stable and robust adaptation.

3.3. VLM-Guided Model Alignment

Building upon the overall framework described above, VGMA focuses on aligning the target domain to the source domain under the guidance of VLMs. However, achieving such alignment is challenging because SFDA has no access to source data and lacks direct semantic evidence. To remove this data dependency, we align target features to the classifier’s discriminative space. Each category is represented by its classifier weight vector, which encodes source trained semantic knowledge. Aligning target features to

these prototypes enables the model to recover class-wise decision boundaries in the target domain without requiring source data. Naive prototype alignment, focuses solely on feature similarity and often ignores the semantic relationships between classes, leading to mismatched mappings under large domain shifts. To achieve semantically consistent alignment, it is essential to couple the feature space and the prediction space during transport.

To capture both feature and semantic consistency, we reformulate the alignment as an OT problem guided by the semantic priors from VLMs. In this problem, the target samples constitute the source distribution to be transported, while the class prototypes constitute the target distribution. Following the principle of Joint Distribution Optimal Transport (JDOT) [4], the transport cost between a target sample x_i^t and a prototype w_j is defined as a joint distance over the feature and prediction spaces

$$C_{i,j}^m = \alpha \cdot d(z_i^m, w_j) + L(q_i^m, y_j), \quad (1)$$

where $z_i^m = f(x_i^t)$ denotes the feature representation of the i -th target sample extracted by the task model, q_i^m is its corresponding class probability vector, w_j represents the classifier weight vector serving as the prototype for class j , and y_j is the one-hot label associated with that prototype. The term $d(\cdot, \cdot)$ measures the cosine distance in the feature space, and $L(\cdot, \cdot)$ denotes the cross-entropy between two class distributions. The coefficient α normalizes the distance term, defined as $\alpha = 1 / \max_{i,j} d(z_i^m, w_j)$.

Given this cost definition, the OT plan Γ^* is obtained by minimizing the total transport cost, and the overall distribution alignment loss is formulated as

$$\mathcal{L}_{\text{Align}} = \sum_{i=1}^{N_t} \sum_{j=1}^K \Gamma_{i,j}^* C_{i,j}^m. \quad (2)$$

This loss drives each target feature toward semantically consistent prototypes, thus aligning the target distribution with source semantics.

Although the above objective provides a principled formulation for feature and prototype alignment, it still suffers from a critical limitation in that it relies solely on the task model's own predictions. Under large domain shifts, such predictions may be unreliable, leading to biased transport couplings and unstable alignment. In this case, the model essentially guides itself without any external semantic reference, which can amplify existing prediction errors. To enhance stability and semantic fidelity, we therefore introduce guidance from VLMs and compute the OT plan under their supervision, leveraging their semantic priors to provide a more reliable alignment reference. Accordingly, the transport computation is reformulated to incorporate these VLM-derived semantic priors. Specifically, while Eq. 1 defines the cost matrix used in the alignment loss, it is no

longer directly used for transport computation. Instead, the cost for plan computation is decoupled from that for model optimization, and a VLM-guided prior is constructed to estimate the transport plan in a semantically grounded space. This separation enables the OT plan to serve as a stable alignment reference even when the task model's predictions are uncertain. Formally, the VLM-guided transport cost between a target sample and a prototype is defined as

$$C_{i,j}^v = \alpha \cdot (1 - q_{i,j}^v) + L(q_i^m, y_j), \quad (3)$$

where $q_{i,j}^v$ denotes the VLM-predicted probability of class j for the target sample i , serving as a semantic prior that reflects their cross-modal affinity. The probability vector q_i^v is derived from the similarity between the VLM's image and text embeddings, computed via inner product followed by softmax normalization

$$q_i^v = \text{softmax}(S_{i,:}^v), \quad S^v = z_{\text{img}} z_{\text{text}}^\top. \quad (4)$$

Here, z_{img} and z_{text} denote the image and text embeddings extracted from VLMs, respectively, and $S_{i,j}^v$ measures the similarity between the i -th target image and the j -th class text prompt. The softmax operation converts these similarities into a normalized probability distribution over classes. This formulation allows a VLM's semantic knowledge to directly modulate the transport plan, guiding the alignment toward semantically consistent couplings across domains. Using this VLM-regularized cost C^v , the OT plan Γ^* is obtained by solving

$$\Gamma^* = \arg \min_{\Gamma \in \Pi(\mu_t, \mu_p)} \int_{(\Omega \times \mathcal{C})^2} C^v d\Gamma, \quad (5)$$

where $\Pi(\mu_t, \mu_p)$ denotes the set of admissible couplings with marginals μ_t and μ_p , representing the target prediction and prototype distributions, respectively.

Based on Eq. 1 and 5, the model's cost matrix and the prior-informed transport plan are obtained and subsequently used in Eq. 2 to iteratively optimize the network parameters. In mini-batch training, class proportions fluctuate. Bias in the prototype marginal used to construct the transport plan can degrade transfer quality and destabilize alignment. To address this issue, μ_t is set to a uniform distribution, indicating that each sample is equally weighted during transport. μ_p is estimated from VLM predictions, reflecting the mini-batch's probability distribution over destination prototypes. Here, $\mu_t(i)$ denotes the marginal over target samples, while $\mu_p(j)$ denotes the marginal over class prototypes. The marginals used for OT are defined as

$$\mu_t(i) = \frac{1}{|B|}, \quad \mu_p(j) = \frac{1}{|B|} \sum_{i \in B} q_{i,j}^v. \quad (6)$$

To stabilize training under SFDA and improve feature discriminability, two auxiliary objectives are introduced in

the VGMA stage. The first is a view-level contrastive loss that enforces augmentation-invariant representations without requiring labels or negative pairs. Given two stochastic augmentations x_1, x_2 of the same target image, features are extracted and projected as

$$\tilde{z}_1^m = h(f(x_1)), \quad \tilde{z}_2^m = h(f(x_2)). \quad (7)$$

where all vectors are ℓ_2 -normalized. Let $\text{sg}(\cdot)$ denote the stop-gradient operator. The contrastive loss minimizes the cosine distance between the two embeddings

$$\begin{aligned} \mathcal{L}_{\text{Con}} = & \frac{1}{2} [1 - \cos(\tilde{z}_1^m, \text{sg}(\tilde{z}_2^m))] \\ & + \frac{1}{2} [1 - \cos(\tilde{z}_2^m, \text{sg}(\tilde{z}_1^m))]. \end{aligned} \quad (8)$$

The second objective, information maximization, sharpens predictions while maintaining balanced class utilization within each batch. Let q_i^m be the model-predicted probability for the i -th target sample in a batch B , and $H(\cdot)$ denotes the entropy function. The loss is expressed as

$$\mathcal{L}_{\text{IM}} = \frac{1}{|B|} \sum_{x_i \in B} H(q_i^m) - H\left(\frac{1}{|B|} \sum_{x_i \in B} q_i^m\right). \quad (9)$$

Together, $\mathcal{L}_{\text{Con}}$ and $\mathcal{L}_{\text{IM}}$ act as regularizers that enhance representation consistency and prediction confidence, leading to more stable adaptation in the source-free setting.

3.4. Model-Guided VLM Adaptation

Although VLMs exhibit strong zero-shot generalization, their guidance to the task model can be suboptimal in the target domain due to the lack of task-specific cues. To remedy this, the MGVA stage distills high-confidence class information from the task model back into VLMs, enabling them to perceive target-domain context and thus provide stronger guidance. Concretely, we insert a lightweight adapter into $\mathcal{V}$ and *only* fine-tune this module, while freezing the task model $\mathcal{G} = (f, c)$.

Given a mini-batch B of $|B|$ target samples, retain the Top- k classes for each sample i to form a sparse matrix S^m

$$S_{i,c}^m = \begin{cases} q_{i,c}^m, & \text{if } c \in \text{Top-}k(q_{i,:}^m), \\ 0, & \text{otherwise,} \end{cases} \quad (10)$$

and renormalize over the retained support to obtain a valid distribution

$$\tilde{q}_i^m = \text{softmax}(S_{i,:}^m). \quad (11)$$

A VLM generates class probabilities q^v as defined in Eq. 4. We then define the model to VLM distillation objective as the average Kullback–Leibler(KL) divergence

$$\mathcal{L}_{\text{VLM}} = \frac{1}{|B|} \sum_{x_i \in B} \text{KL}(\tilde{\mathbf{q}}_i^m \parallel \mathbf{q}_i^v). \quad (12)$$

This objective aligns VLMs with the task model’s confident predictions while avoiding hard-label noise, thereby reducing the distributional ambiguity introduced by holistic alignment. This makes $\mathcal{V}$ more target-aware and strengthens its guidance for subsequent adaptation.

3.5. Overall Objective

To coordinate the two stages and drive joint progress of a VLM and the task model, our training framework adopts an alternating two-stage optimization strategy that forms a bidirectional feedback loop between them. In the VGMA stage, only the task model is optimized with a frozen VLM, using the alignment loss and two auxiliary objectives

$$\mathcal{L}_1 = \mathcal{L}_{\text{Align}} + \mathcal{L}_{\text{Con}} + \mathcal{L}_{\text{IM}}. \quad (13)$$

This stage emphasizes distribution alignment under a VLM’s semantic guidance, encouraging the task model to learn target-domain features that are both discriminative and semantically consistent. In the MGVA stage, only a VLM is refined via task-specific distillation while the task model remains frozen, and the overall objective is given by

$$\mathcal{L}_2 = \mathcal{L}_{\text{VLM}}. \quad (14)$$

Through alternating updates, the two stages interact in a mutually corrective manner, ensuring stable optimization and improving final performance.

4. Experiments

4.1. Datasets

VSFOT is evaluated on four widely used benchmarks to demonstrate its effectiveness across diverse transfer scenarios. **Office-31** [30] is a classic domain adaptation dataset comprising 4,110 images from three domains: Amazon (A), Webcam (W), and DSLR (D), with 31 shared classes. **Office-Home** [38] contains about 15,500 images in four domains: Art (Ar), Clipart (Cl), Product (Pr), and Real World (Rw), covering 65 classes. **VisDA** [25] is a large-scale simulation to reality benchmark. It consists of over 280K images across 12 classes, with the source domain being synthetic renderings and the target domain being real world photographs. **DomainNet-126** [26] is a subset of the DomainNet dataset. It contains four parts: Real (R), Sketch (S), Clipart (C), Painting (P), covering 126 classes. We evaluate 6, 12, 1, and 12 adaptation tasks on Office-31, Office-Home, VisDA, and DomainNet-126 respectively.

4.2. Competitors

We compare the proposed method with three classes of approaches. First, the original task model and the VLM model are included as baselines to illustrate the impact of distribution shift. Second, comparisons are made with representative SFDA methods. Specifically, the following methods

are considered: SHOT [15], NRC [41], AaD [43], AdaCon [2], CoWA [11], PLUE [17], TPDS [34], DIFO [35], and ProDe [36], where DIFO and ProDe are VLM-based approaches. Third, UDA methods that leverage VLMs are included. These typically assume access to source data during training and include DAPL [7], PADCLIP [10], ADCLIP [32], PDA [1], and DAMP [6].

4.3. Implementation Details

Following SHOT [15], ResNet50 [8] is adopted as the backbone for Office-31, Office-Home, and DomainNet-126, while ResNet101 is used for VisDA. All models are initialized with ImageNet-1K [5] pretrained weights. During the adaptation process, CLIP is employed as the VLM, with lightweight adapters inserted into the last three blocks [29]. Each adapter consists of two linear layers with a ReLU activation and a hidden dimension of 64. The OT problem is addressed using the Sinkhorn algorithm, with the entropy regularization coefficient set to 0.2 for all datasets. All results are reported as the average over three runs.

4.4. Experimental Results

4.4.1. Comparison on Domain Adaptation Setting

Tables 1, 2, and 3 summarize the performance of VSFOT across four benchmark datasets. For DomainNet-126, Top-1 accuracy is reported for each transfer task, while average accuracy is used for other datasets. As shown in Tables 1 and 3, VSFOT consistently outperforms existing baselines on Office-31 and Office-Home. Specifically, VSFOT achieves gains of 0.09% and 1.14% over the second-best methods on Office-31 and Office-Home, respectively. On the more challenging DomainNet-126 (see Table 2), VSFOT surpasses the DIFO by a substantial margin of 2.55%. For VisDA (see Table 3), VSFOT ranks second among all compared methods. The performance drop is largely attributed to the absence of an error correction mechanism, which allows confusion in a few classes to mislead the model during training. Nevertheless, even under such a challenging setting, VSFOT still demonstrates competitive performance. In summary, VSFOT maintains strong performance across all benchmarks. It outperforms all conventional SFDA methods, highlighting the essential role of the VLM in enhancing domain adaptation. Compared to recent VLM-based UDA approaches, VSFOT achieves comparable or even superior results, indicating its potential as a more challenging yet efficient solution for domain adaptation. These results collectively validate the strong cross-domain generalization capability of VSFOT in domain adaptation.

4.4.2. Compare with CLIP Zero-shot Predictions

To evaluate the performance gains brought by incorporating VLMs into SFDA and to validate the effectiveness of

Method	SF	→A	→D	→W	Avg.
Source	-	61.75	85.95	81.90	76.53
SHOT	✓	72.10	97.10	95.15	88.12
NRC	✓	76.15	97.12	95.90	89.72
AaD	✓	74.12	96.20	95.40	88.57
AdaCon	✓	75.51	85.25	87.20	82.65
CoWA	✓	78.92	95.64	95.23	89.93
PLUE	✓	74.50	94.25	93.12	87.29
TPDS	✓	75.80	96.60	95.20	89.20
DIFO	✓	83.00	98.00	96.35	92.45
ProDe	✓	82.80	98.20	96.70	92.57
VSFOT	✓	83.42	98.26	96.29	92.66

Table 1. Top-1 accuracy (%) on Office-31. SF indicates whether the method follows the source-free setting. Bold indicates the best accuracy.

the VLM-guided adaptation, we compare the original CLIP model with our proposed VSFOT. Table 4 reports the average accuracy across four datasets, where the results of VSFOT are computed by averaging the outcomes on each task. Although VLMs demonstrate strong zero-shot capabilities, VSFOT achieves consistently higher accuracy. Specifically, compared to zero-shot predictions from CLIP, VSFOT achieves improvements of 12.55%, 6.3%, 5.96%, and 4.34% on the four datasets, respectively. These results highlight that VSFOT can effectively leverage the knowledge embedded in VLMs to further enhance domain adaptation performance on the target domain.

4.5. Component Analysis

4.5.1. Ablation Study

Table 5 reports the ablation results of various VSFOT modules on Office-Home and DomainNet-126. The first row shows the performance of the source model, where accuracy remains low due to domain discrepancies. Subsequently, introducing $\mathcal{L}_{\text{Align}}$ significantly improves the average accuracy. It brings gains of 21.54% and 20.59% on the two datasets. This improvement comes from combining rich semantic priors from the large model with task-specific knowledge from the task model. Moreover, incorporating $\mathcal{L}_{\text{VLM}}$ further enhances performance by 2.71% and 1.25%, validating the effectiveness of model-to-VLM knowledge transfer in strengthening the VLM’s guidance. Finally, adding two additional loss terms improves feature representation and class discriminability, yielding respective gains of 1.17% and 0.19%.

In addition, Table 5 investigates the impact of marginal distribution estimation and prior-guided adaptation. Marginal estimation is key to VSFOT, supplying a reliable mini-batch target distribution. To expose transport bias from abnormal marginals, we fix the class marginal to uniform and validate the module’s effectiveness. Ablation results show that removing marginal estimation leads

Method	SF	C→P	C→R	C→S	P→C	P→R	P→S	R→C	R→P	R→S	S→C	S→P	S→R	Avg.
Source	-	51.30	67.70	52.00	58.10	78.80	54.50	59.60	66.70	53.20	60.60	57.30	67.40	60.60
SHOT	✓	64.47	79.03	58.79	67.76	81.78	61.43	66.95	67.99	56.93	70.01	64.11	77.94	68.10
NRC	✓	62.08	77.74	59.26	62.40	81.20	59.83	64.06	70.28	59.43	69.03	65.62	78.24	67.43
AdaCon	✓	60.20	74.94	55.44	61.28	78.59	58.78	63.80	68.57	55.03	66.34	65.37	75.67	65.33
CoWA	✓	63.73	80.03	60.30	66.80	80.69	61.56	68.86	68.01	60.86	69.24	66.23	79.78	68.84
PLUE	✓	60.38	73.34	55.51	62.32	79.39	57.22	61.77	66.80	54.03	68.40	64.48	75.54	64.93
TPDS	✓	62.90	76.26	60.75	64.69	79.87	61.97	67.27	67.26	58.50	68.19	64.85	75.39	67.32
DAPL	×	72.76	87.65	66.48	73.27	86.71	65.75	74.16	72.91	66.02	74.75	73.34	87.22	75.08
ADCLIP	×	71.57	88.59	66.37	72.77	86.38	65.50	73.76	72.29	67.93	72.24	73.82	90.25	75.12
DAMP	×	76.09	89.48	71.53	75.00	89.02	69.89	73.73	74.83	70.09	74.49	75.37	88.31	77.32
DIFO	✓	77.53	86.51	74.90	79.60	87.19	75.80	81.74	76.97	75.99	80.61	76.68	87.43	80.08
VSFOT	✓	79.48	90.40	77.21	83.02	90.54	77.46	83.30	79.76	76.92	83.45	79.64	90.37	82.63

Table 2. Top-1 accuracy (%) on DomainNet-126. SF denotes source-free setting. Bold indicates the best accuracy.

Method	SF	Office-Home					VisDA
		→Ar	→Cl	→Pr	→Rw	Avg.	→Real
Source	-	57.73	41.00	70.80	71.73	59.92	51.20
SHOT	✓	70.77	57.49	80.58	79.82	72.16	84.85
NRC	✓	67.79	57.88	81.95	81.12	72.18	86.13
AaD	✓	68.64	58.93	81.23	80.89	72.42	89.17
AdaCon	✓	65.14	48.45	79.50	76.03	67.28	85.61
CoWA	✓	70.33	58.11	81.31	80.96	72.68	87.08
PLUE	✓	69.63	57.72	77.09	80.95	71.35	89.42
TPDS	✓	72.43	58.21	80.84	82.21	73.42	86.24
DAPL	×	74.45	53.87	83.69	84.23	74.06	88.37
PADCLIP	×	78.21	59.09	86.24	85.14	77.17	89.21
ADCLIP	×	77.38	55.06	86.07	84.99	75.88	88.32
PDA	×	73.73	55.88	85.80	86.27	75.42	85.28
DAMP	×	76.59	60.62	88.64	87.11	78.24	89.20
DIFO	✓	81.68	70.37	90.81	88.72	82.90	90.33
ProDe	✓	82.83	72.66	91.31	90.02	84.20	91.04
VSFOT	✓	83.39	74.76	91.80	91.40	85.34	90.57

Table 3. Top-1 accuracy (%) on Office-Home and VisDA. SF indicates source-free setting. Bold indicates the best accuracy.

Method	Office-31	Office-Home	VisDA	DomainNet-126
CLIP	80.11	79.04	84.61	78.29
VSFOT	92.66	85.34	90.57	82.63

Table 4. Average Top-1 accuracy (%) comparison between the original VLM and VSFOT. Bold indicates the best accuracy.

to performance drops of 11.72% and 18.29%, indicating that the uniform distribution assumption severely harms adaptation accuracy. For the prior guidance part, we replace Eq. 3 with its original form Eq. 1, preventing the VLM from participating in the distribution alignment process. In this case, the SFDA process reduces to aligning target features to the source distribution purely based on OT. Ablation results show that relying solely on OT and the model’s self-alignment capability fails to achieve satisfactory performance. However, it still provides certain

$\mathcal{L}_{\text{Align}}$	$\mathcal{L}_{\text{VLM}}$	$\mathcal{L}_{\text{Con}}$	$\mathcal{L}_{\text{IM}}$	Office-Home	DomainNet-126
				59.92	60.60
✓				81.46	81.19
✓	✓			84.17	82.44
✓	✓	✓		85.19	82.52
✓	✓	✓	✓	85.34	82.63
w/o Marginal Estimation				73.62	64.34
w/o Prior Guidance				74.13	64.19

Table 5. Ablation results on Office-Home and DomainNet-126. Average Top-1 accuracy (%) is reported to assess the effect of each component. Bold indicates the best accuracy.

improvements, outperforming the only source baseline by 14.21% on Office-Home and 3.59% on DomainNet-126.

4.5.2. Effectiveness of the Distribution Alignment Mechanism

To assess the effectiveness of our distribution alignment strategy, we conduct experiments on Office-Home and DomainNet-126 by replacing key components in our method. We replace the loss terms in VSFOT with soft-label alignment and hard-label supervision to validate the rationality of our design. Specifically, KL divergence is adopted for soft-label alignment, while cross-entropy loss is applied to hard-label supervision using Top-1 pseudo labels. Table 6 reports the four designed variants and their results. Case 1 uses KL divergence as loss to replace the proposed $\mathcal{L}_{\text{Align}}$. The results show that KL divergence fails to facilitate effective knowledge transfer, and its performance even falls short of the VLM’s zero-shot predictions. Case 2 and 3 respectively incorporate soft-label and hard-label in place of VSFOT’s corresponding components. The results indicate that the $\mathcal{L}_{\text{VLM}}$ used in VSFOT effectively mitigates the ambiguity caused by KL based distillation and the noise sensitivity of hard-label supervision. Case 4 shows inferior performance, as it employs a bidirectional distillation scheme based on KL divergence. Due to the limited capac-

Setting	$\mathcal{G} \rightarrow \mathcal{V}$	$\mathcal{V} \rightarrow \mathcal{G}$	Office-Home	DomainNet-126
Source	N/A	N/A	59.92	60.60
CLIP	N/A	N/A	79.04	78.29
Case. 1	Soft	$\mathcal{L}_{\text{VLM}}$	64.90	61.15
Case. 2	$\mathcal{L}_{\text{Align}}$	Soft	75.41	78.46
Case. 3	$\mathcal{L}_{\text{Align}}$	Hard	85.09	82.01
Case. 4	Soft	Soft	64.88	61.15
VSFOT	$\mathcal{L}_{\text{Align}}$	$\mathcal{L}_{\text{VLM}}$	85.34	82.63

Table 6. Comparison of different distribution alignment strategies. The table shows average Top-1 accuracy (%) on Office-Home and DomainNet-126. $\rightarrow$ denotes the distillation direction. N/A means not applicable. Soft and Hard indicate alignment types replacing the loss terms. Bold indicates the best accuracy.

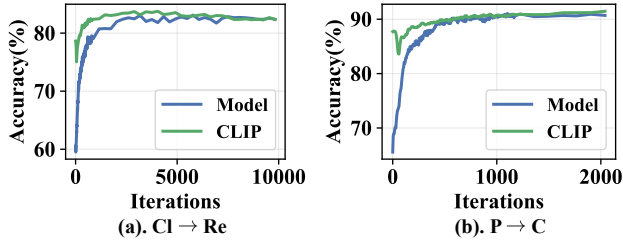


Figure 3. Average Top-1 accuracy of the VLM and task model on $\text{CI} \rightarrow \text{Re}$ from Office-Home and $\text{P} \rightarrow \text{C}$ from DomainNet-126.

ity of KL divergence to transfer rich semantic knowledge, this setting fails to yield competitive results.

4.5.3. Analysis of the Bidirectional Distillation

To investigate how the proposed bidirectional distillation mechanism enhances the VLM’s capacity to guide the task model during SFDA, we track the Top-1 accuracy result of both the task model and the VLM throughout training. Figure 3 shows the results on task $\text{P} \rightarrow \text{C}$ using DomainNet-126 and task $\text{CI} \rightarrow \text{Re}$ using Office-Home. Notably, both the task model and the VLM exhibit consistent accuracy improvements and eventually converge to similar performance levels. These results indicate that the bidirectional distillation mechanism effectively transfers general knowledge from the VLM to the task model, while simultaneously propagating task-specific knowledge from the task model back to the VLM. Through mutual refinement, both models co-evolve, allowing the task model to leverage task-specific guidance and surpass the original VLM.

4.6. Hyperparameter Analysis

We analyze key hyperparameters and report average Top-1 accuracy across multiple datasets. Figure 4(a) presents the effect of adapter depth on Office-Home. A small number of layers reduces capacity due to limited trainable parameters. Using about three adapters yields nearly the best and most stable performance, and adding additional adapters offers only marginal benefit, indicating a capacity limit of the

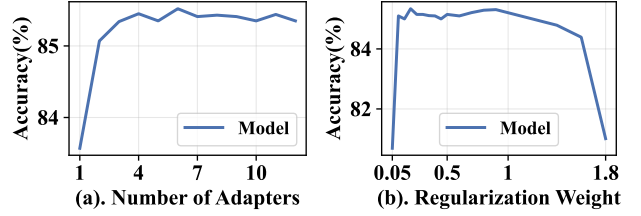


Figure 4. Average Top-1 accuracy on Office-Home under different adapter numbers and OT regularization weights.

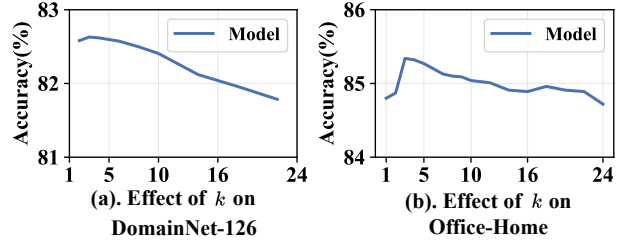


Figure 5. Average Top-1 accuracy on DomainNet-126 and Office-Home with different k values.

approach. Figure 4(b) reports the effect of the Sinkhorn entropy regularization weight on Office-Home. Extremely small or large values degrade performance, whereas an intermediate range balances sparsity and smoothness in the OT plan and delivers stable, near-optimal results. Finally, we evaluate different top- k settings in the MGVA stage on Office-Home and DomainNet-126, as shown in Figure 5. The parameter k determines the number of masked classes. Large k introduces noise and weakens alignment, while small k may exclude correct classes. The results show a non-monotonic dependence on k , with an optimum at $k = 3$ that yields the highest and most stable performance across both datasets.

5. Conclusion

We introduce VSFOT, a SFDA method that taps a pretrained VLM’s semantic priors via bidirectional distillation. Instead of relying on pseudo labels, VSFOT uses the VLM-guided OT to align feature distributions, reducing noise. It also distills target-domain signals back into the VLM, giving it task-specific knowledge and stronger contextual sensitivity. On multiple SFDA benchmarks, VSFOT consistently shows higher accuracy and robustness under large domain shifts. These results highlight the value of integrating VLMs into SFDA and using OT for semantic alignment, encouraging further work on semantically guided, distribution-aware adaptation in source-free settings.

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